MIL-C-46200A (MU)
2 JULY 1963
SUPERSEDING
MIL-C-46200 (Ord)
31 MAY 1960

MILITARY SPECIFICATION

CARTRIDGE, THRUSTER, M42A1

1. SCOPE

1.1 This specification covers the Cartridge, Thruster, M42A1 for use with propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons, General Specification for.

MIL-E-5272 — Environmental Testing, Aeronautical and Associated Equipment, General Specification for.

MIL-I-45208 — Inspection Requirements, General Specification for.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-639 — Surface Roughness Values Preferred for Army Ordnance Materiel.

DRAWINGS

MUNITIONS COMMAND

D8593889 — Cartridge, Thruster, M42A1, Assembly.

F8797435 — Carton, Packing, Ammunition, for Cartridge, Thruster, M42A1.

F8797436 — Box, Packing, Ammunition, for Cartridge, Thruster, M42A1.

IEL-8593889 — Inspection Equipment List for Cartridge, Thruster, M42A1.

PUBLICATIONS

Frankford Arsenal Manual for Functional Tests — Cartridge Actuated Devices for Aircraft Ejections Systems.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1375

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:12Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 General. Materials, parts, assemblies, physical and chemical requirements shall conform to the applicable drawings, referenced specifications and the following:

3.2 Initial production sample. Unless otherwise directed by the contracting officer, an initial production sample shall be required.

3.3 Protective finish. When protective finish is required by component details, finish requirements shall be as required on the drawings and referenced specification and shall apply to all surfaces of the assembled cartridge regardless of contour configuration.

3.4 Primer. The 72M Primer shall be used.

3.5 Cartridge assembly. Components shall be combined to produce an assembly which conforms to the requirements of Drawing D8593889.

3.5.1 *Vibration.* The assembled cartridge shall withstand vibration prescribed in Specification MIL–E–5272, Procedure XII, without evidence of any disintegration of the primer, igniter or propellant or derangement of components.

3.6 Functional requirements.

3.6.1 The cartridge shall not fail to fire.

3.6.2 *Powder charges.* The igniter and propellant charges shall be fixed by the contractor to produce the specified ballistic requirements (see 6.1.)

3.6.3 *Functioning.* The cartridge, when tested in an M1A1 Thruster, temperature conditioned at minus 65 degrees Fahrenheit (F) to minus 70 degrees F and plus 160 degrees F to plus 165 degrees F, shall produce sufficient thrust while propelling a 20 pound mass along a horizontal path to overcome the following restraining forces:

1000 pounds at zero inches stroke

6000 pounds at 0.25 inches stroke

3200 pounds at 0.40 inches stroke

1000 pounds at 0.75 inches stroke

3.7 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.7.1 *Metal defects.* All metal parts shall be free from cracks, splits, laminations, perforations, porosity or any similar defect.

3.7.2 *Surface finish.* The requirements for surface finish are as detailed on the applicable drawing and referenced military standard.

3.7.3 *Burr.* No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.4 *Foreign matter.* No part or assembly shall contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Quality assurance terms and definitions.* Reference shall be made to Standard MIL–STD–109 to define quality assurance terms used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:12Z
Check the source to verify that this is the current version before use.

4.1.2 *Inspection.* Inspection shall be in accordance with Specifications MIL–A–2550 and MIL–I–45208, except as specified herein.

4.2 Initial production sample. Prior to the initiation of regular production, the contractor shall submit an initial production sample to a designated Government approved facility for evaluation. The sample shall consist of 30 cartridges and 10 sets of unassembled components including explosive components. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, must be the same as will be used in regular production and shall be obtained from the same source of supply. The contractor shall inspect the sample in accordance with his inspection system and other inspection requirements of Specification MIL–I–45208. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples unit such time as an acceptable sample is submitted.

4.3 Inspection provisions.

4.3.1 *Lot.*

4.3.1.1 *Presentation of lots.* The product shall be presented in accordance with Standard MIL–STD–105.

4.3.1.2 *Size of lot.* Unless otherwise specified by the procuring activity, an ammunition lot shall consist of 600 cartridges ,maximum, exclusive of test samples and shall consist of:

(a) Components produced by a homogeneous process.

(b) Propellant from not more than one lot.

(c) Igniter from not more than one lot.

(d) Primers from not more than one lot.

4.3.2 *Examination.* One hundred percent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plan and criteria in Tables I and II. For an individual defect (single characteristic), the criteria in Table I shall be used. For total defects within a major or minor group, the criteria in Table II shall be used. All non-conforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter. The Ordnance Standard Finish Blocks prescribed in Standard MIL–STD–639 shall be used as a basis of comparison for surface roughness determination.

TABLE I

Lot size	*Sample size*	*Acceptance numbers*	
		Major	*Minor*
1 – 300	100% examination	0	0
301 – 1300	110	1	2
1301 – 3200	150	2	3

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:12Z
Check the source to verify that this is the current version before use.

TABLE II

No. of defects in class	*Total major defects allowable*
2 – 3	1
4 – 7	2
8 – 19	3

No. of defects in class	*Total minor defects allowable*
2 – 3	2
4 – 7	3
8 – 16	4

4.3.2.1 *Classification of defects.*

4.3.2.1.1 *Cartridge, Thruster, M42A1, Assembly, Drawing D8593889.*

Defect	*Method of inspection*
Critical	
1. Depth to primer pocket face	Gage
2. Primer seating depth	Gage
3. Any part missing	X-ray/Visual [1]
4. Improper assembly	X-ray/Visual [1]
5. Mixed ammunition	Visual
6. Metal defective	Visual
7. Crimp missing or incorrect	Visual
8. Marking missing, incorrect or illegible	Visual
Major	
101. Cartridge profile failure	Gage
102. Diameter of body, max.	Gage
103. Diameter of head, max.	Gage
104. Length to datum diameter, max.	Gage
105. Primer crimp improper	Visual
106. Protective finish inadequate or defective	Visual
107. Foreign matter	Visual
108. Burr	Visual
Minor	
201. Total length, max.	Gage
202. Dent	Visual
203. Draw scratch	Visual
204. Fold, wrinkle, buckle or bulge	Visual
205. Varnish inadequate or missing	Visual
206. Surface finish improper	Visual

[1] The assembled cartridge shall be x-rayed with the long axis of the cartridge perpendicular to the rays of the x-ray machine. The x-ray negatives, properly identified, shall be forwarded to the procuring activity.

4.3.2.1.2 *Primer, Percussion, 72M, Drawing C7645332 Prior to Assembly Into a Cartridge.*

Defect	*Method of inspection*
Critical	
1. Improper assembly	X-ray [2]
2. Any part missing	X-ray [2]
3. Mixed primers	Visual
4. Primer pellet broken, cracked or dislodged	Visual
5. Primer foil discolored	Visual
6. Metal defective	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:12Z
Check the source to verify that this is the current version before use.

Defect	Method of inspection
Major	
101. Burr	Visual
102. Foreign matter	Visual
Minor	
None defined.	

[3] Prior to assembly into a cartridge, the primer shall be x-rayed with the side of the cup perpendicular to the rays to determine compliance with the requirements of the applicable drawing. The x-ray negatives, properly identified, shall be forwarded to the procuring activity.

4.3.2.1.3 *Carton, Packing, Ammunition, for Cartridge, Thruster, M42A1, Drawing F8797435.*

Defect	Method of inspection
Critical	
None defined.	
Major	
None defined.	
Minor	
201. Contents loose	Manual/Visual
202. Quantity of contents incorrect	Visual
203. Heat sealer incomplete, loose or distorted	Visual
204. Separator or filler missing	Visual
205. Cut or damage through impregnated box	Visual
206. Marking, missing, incorrect or illegible	Visual

4.3.2.1.4 *Box, Packing, Ammunition, for Cartridge, Thruster, M42A1, Drawing F8797436.*

Defect	Method of inspection
Critical	
None defined.	
Major	
None defined.	
Minor	
201. Contents loose	Manual/Visual
202. Hardware missing, loose, broken or improperly engaged	Visual
203. Strapping missing; loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of cartons incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.3.2.2 *Primer examination.* Prior to assembly into a cartridge, each primer shall be visually examined for any discoloration of the foiling paper or a malformation of any kind. Any discoloration of the foiling paper or malformation shall be cause for rejection of the primer.

4.3.3 *Inspection testing.* The following tests shall be performed in accordance with the provisions of 4.4, except as provided in Specification MIL–I–45208.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:12Z
Check the source to verify that this is the current version before use.

4.3.3.1 *Protective finish.* The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.3.3.2 *Airtightness.* Each cartridge shall be subjected to the specified pressure differential. Failure to comply with the requirements of the drawing shall be cause for rejection of the cartridge.

4.3.3.3 *Vibration test — Initial production sample.* Twelve of the 30 assembled x-rayed cartridges of the initial production sample, properly identified, shall be subjected to vibration as directed in 3.5.1. After the vibration test, the 12 vibrated cartridges shall be x-rayed as directed in Note [1] of 4.3.2.1.1. Evidence of any disintegration of the primer or propellant or derangement of any component shall be cause for rejection of the lot.

4.3.3.3.1 *Vibration test — Production lot sample.* When it becomes necessary to make a change in the lot of primers, propellant or igniter after production is initiated, 30 cartridges from the first production lot thereafter shall be tested in accordance with 4.3.3.3.

4.3.3.4 *Functioning.* Thirty assembled cartridges, from the initial production sample and ten cartridges from each subsequent production lot shall be functionally tested. Failure of any cartridge to comply with the requirements of 3.6 shall be cause for rejection of the lot.

4.3.4 *Inspection equipment.* The examination and tests shall be made using equipment listed on IEL–8593889, except as provided in Specification MIL–I–45208.

4.4 Test methods and procedures.

4.4.1 *Protective finish.* Test procedures for protective finish shall be in accordance with the applicable specification referenced on the drawing.

4.4.2 *Airtightness.* The method for determining airtightness shall be by use of equipment approved by the procuring activity.

4.4.3 *Vibration.* The method and procedure for performing the vibration test shall be in accordance with Specification MIL–E–5272, Procedure XII.

4.4.4 *Functional tests.* The following tests will be performed by the Government:

4.4.4.1 *Initial production sample.* Cartridges of the initial production sample (including the twelve vibrated cartridges) shall be grouped with 6 of the vibrated cartridges in each group, temperature conditioned and tested as follows: Fifteen cartridges conditioned at minus 65 degrees to minus 70 degrees F; fifteen cartridges conditioned at plus 160 degrees to plus 165 degrees F and tested in accordance with 4.4.4.4.

4.4.4.2 *Production lot sample.* Cartridges of the production lot sample shall be grouped, temperature conditioned as follows: Five cartridges conditioned at minus 65 degrees F to minus 70 degrees F, five cartridges at plus 160 degrees F to plus 165 degrees F, and then tested in accordance with 4.4.4.4.

4.4.4.3 *Temperature conditioning.* Cartridges shall be loaded into acceptable M1A1 Thrusters with end caps to cover the gas ports during the conditioning period. The temperatures of the propellant within the cartridges shall be considered the conditioning temperature and will be determined by means of a thermocouple attached to the center of the propellant charge of a specimen cartridge assembled in a standard test thruster device similar to the thrusters undergoing tests. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element con-

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:12Z
Check the source to verify that this is the current version before use.

tained in the specimen cartridge to the required temperature.

4.4.4.4 *Test procedures.* Upon completion of the required temperature conditioning the sample shall be ballistically tested in accordance with "Frankford Arsenal Manual for Functional Tests — Cartridge Actuated Devices for Aircraft Ejection Systems" (or such method as is approved by the contracting officer) within five minutes after removal from the conditioning chamber. The sample shall be reconditioned if this time is exceeded.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking. Packing and marking shall be in accordance with Drawings F8797435 and F8797436 or required by contract.

5.2 Data cards. Data card information shall be in accordance with Specification MIL–A–2550.

6. NOTES

6.1 Advisory propellant loading data.

Black Powder, Class A–4.	**20.0 grains minus 2.0 grains**
Propellant, Class M–10.	**31.0 grains minus 3.0 grains**

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Army—MU

Preparing activity:
Army—MU
Project No. 1375–A546

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:12Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:12Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:12Z
Check the source to verify that this is the current version before use.